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Foundations of Data Science

Background and Research Question

Every year in the United States of America there are approximately 38,000 fatalities from vehicular accidents and over 6 million non-fatal accidents involving passenger cars.[1] This makes vehicles one of the most potent non-disease or crime related dangers affecting the American public. [2] These are not new revelations. For years many organizations have been running campaigns to help make the public more aware of ways to reduce people's chances of being involved in a car accident. Typically the areas of focus are around drivers' specific actions leading up to or while driving, most notably, speeding while driving, driving while under the influence of substances and texting while driving.

However, all previously mentioned factors are related to one's own ability to maintain control of the vehicle they are operating, without consideration of the ability of other drivers to maintain control of their vehicles. To illustrate the difference consider driving in snowy or rainy conditions. This scenario poses an increased risk to a driver both because it is more difficult for a driver to control their car and because other drivers may lose control causing a multiple car car accident. There are two obvious ways to decrease the risk in this scenario, train yourself to handle the adverse conditions, or limit exposure to the adverse conditions. While it is seemingly obvious that weather is an adverse condition, my goal with this project is to further investigate this and other factors that may correlate with car crashes to better understand what causes car crashes. Specifically, I will investigate three categories, times, locations, and weather conditions.

[1] https://www.ddlawtampa.com/resources/.
[2] https://www.cdc.gov/nchs/fastats/injury.htm

The Data Sets

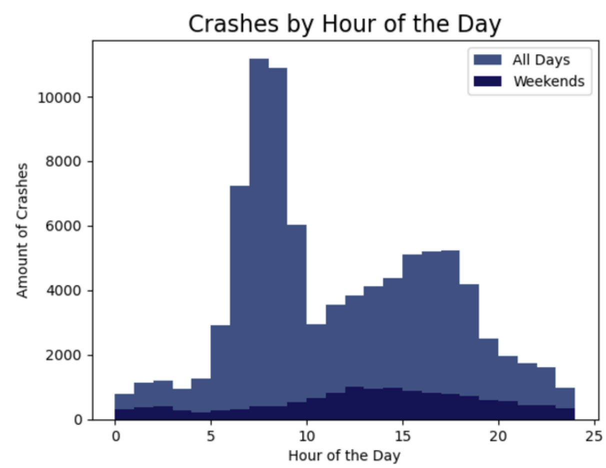
The main dataset that will be used for this project is a collection of data on over 7.7 million car accidents from 2016-2023. The dataset includes specific location information for the accident including latitude, longitude coordinates, county, zip code, etc, weather conditions at the time and place of the accident, and severity of the accident. To perform the investigations I'm interested in, I will need two additional auxiliary data sets. The first will be information on public schools in the United States and its territories, this information is available through nces.ed.gov. The second will be random weather points collected from the relevant locations and times. This data is available through an API provided by open-meteo.com. Using a python script (created with help from ChatGPT) I collected approximately 60,000 data points. The script inputs random latitude and longitude coordinates, bounded by Colorado's borders, and random time coordinates bounded by the minimum and maximum dates from the car crash data set.

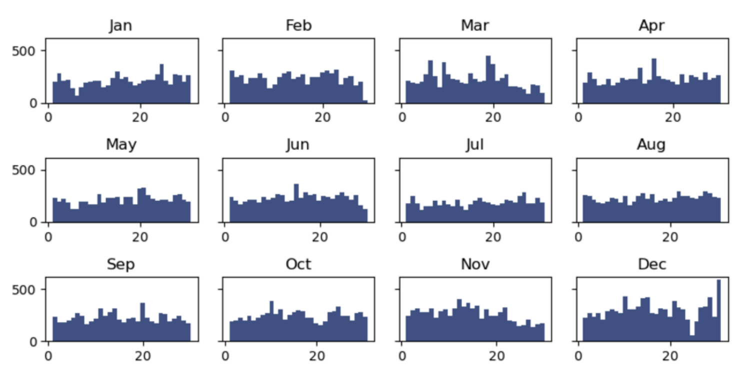
Initial Data Cleaning

The first step to working with any of this data is to perform an initial cleaning that allows me to work with the data in the ED Stem environment. This means the 7.7 million rows and 46 columns from the initial data crash data set need to be cut down significantly. To do this, I am working with data just from Colorado. This brings the data set down to approximately 90,000 rows. Additionally, most of the columns are unnecessary. I only need to keep the coordinates of the crash, and the weather columns that I can compare to data from the API. This leaves me with columns for latitude, longitude, temperature, wind chill, humidity, wind speed, and precipitation. While the school data set is not as bulky, I still only need a subset of the data, highschools in Colorado. One of the columns in the original dataset has state so taking this subset is simple. To take just the highschools I check each row to see if the name has the word "high" in it. While this filter is imperfect, it is good enough for this investigation. Accidentally being too strict and removing high schools without the word “high” does not affect the investigation, and the few non-high schools that may have the word “high” in their name is a relatively small amount that shouldn’t affect the investigation either. Finally, the data from the weather API does not need an initial cleaning since I am collecting only what is necessary to begin with. More specific data cleaning and reformatting is done later on and will be mentioned where relevant.

Investigation 1 - Dangerous Times

The first investigation is regarding when crashes happen. In other words, are there times when crashes are more prevalent so we can avoid driving then. I explore both times during the day and times throughout the year in similar ways; use a histogram to quickly visualize the data, hypothesize, check. From the initial results from the time of day histogram the two major spikes can be possibly attributed to two causes, rush hour or detrimental twilight lighting. These variables can be isolated by investigating weekends. The result is that the spikes completely disappear (see right). Furthermore, weekends are significantly underrepresented. The other time investigation, regarding time of year, revealed that the months have relatively similar amounts of crashes, or atleast I do not see an obvious trend. (Y axis is day of month instead of hour of day here). However this did



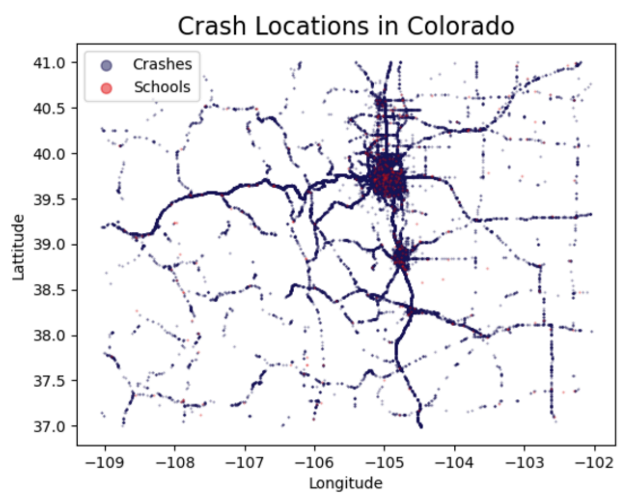


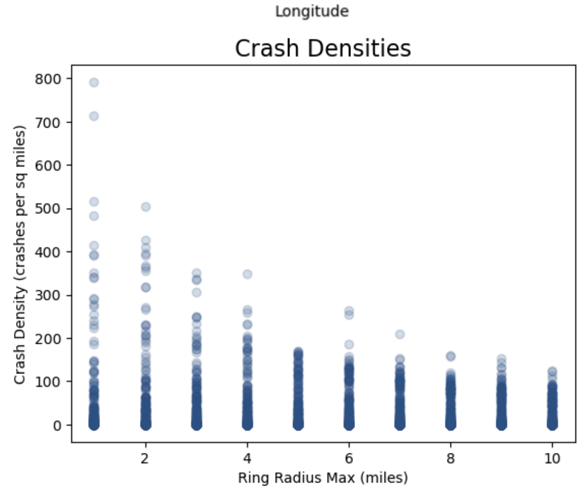
help me notice that the calendar dates with the greatest and fewest (excluding 2/29) crashes are both in December, less than a week apart. Across the dataset, there were only 59 crashes on Christmas (12/5) and there were 589 crashes on New Year's Eve (12/31).

The pattern both of these time investigations reveal is unsurprisingly that when there are more people driving there is a significant increase in the amount of crashes. What is interesting is how little other factors affect the trends. Before the analysis I expected winter months to have more crashes overall and that sunlight would cause more crashes, but the data reveals otherwise.

Investigation 2 - Dangerous Locations

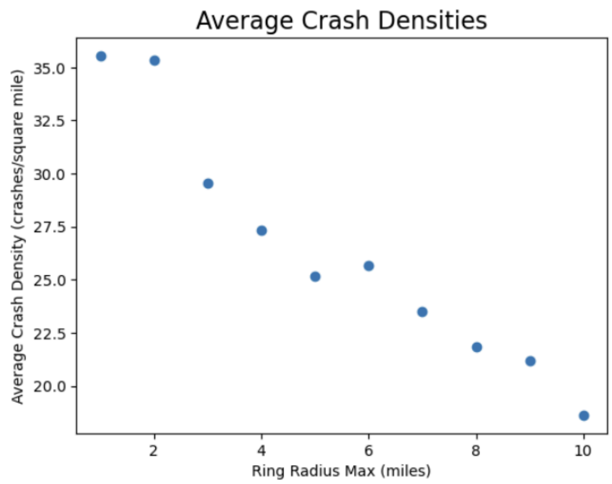
The second investigation was based on the initial hypothesis that because younger drivers are less experienced, there would be more crashes near high schools. The methodology used here could also be applied to investigate other locations such as sports venues or bars. The basic idea is that for every school the distance to each crash can be calculated by accounting for the curvature of earth then using the distance formula. The further from the equator the latitude is, the longitude lines get closer together. Therefore, the distance between longitude lines is: (Distance at the equator) * cos (latitude), and the distance formula can be used to calculate the distance between the schools and the crashes. From here a new 3D data frame is created that will store boolean values for whether the distance is within a certain range. This makes it easy to then calculate





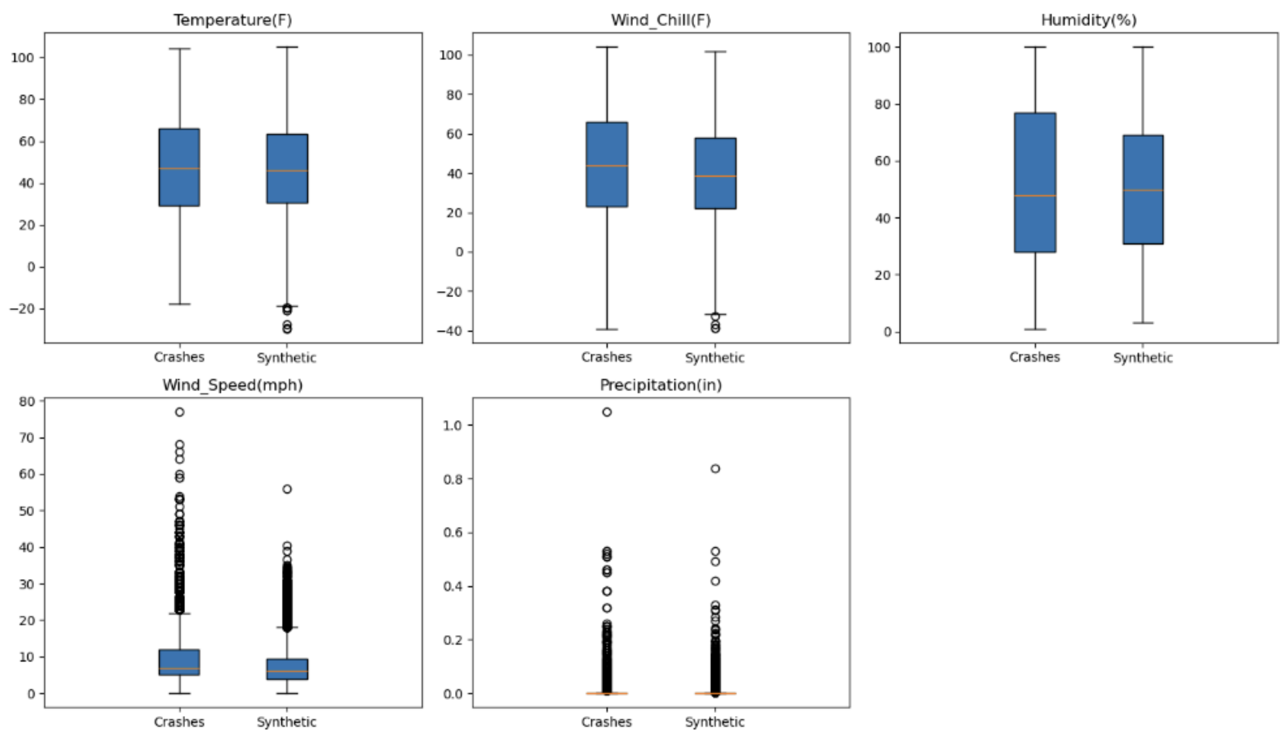
how many crashes happen per range and since each range is a ring the area is simple to calculate and therefore so is the crash density for each school per ring. This can then be plotted (see right), and the mean value for each radius can also be calculated.

This reveals that there is a correlation between proximity to school and how many crashes there are. However, this does not necessarily demonstrate that the cause I assumed is correct. While I believe the method of checking crash density provides a useful insight, the study needs to be adapted to better understand if there are more crashes because the school is there or if schools are typically built near places with many crashes. Given the current data this would be difficult to do, however if data was provided regarding schools opening or closing then crash densities could be compared from before and after the closing or opening of the school. This could also be accomplished by comparing certain road features that are similar, but one is near a school and one is far. For example if there are similar highway exit ramps but some are near schools and some are far, then the amount of crashes at the different exits could be compared.



Investigation 3 - Weather Conditions

The third and final investigation is into the effect that weather conditions have on car crashes. The biggest issue given the initial data set is that it only provides positive cases. In other words, I only have data on when crashes occur, but nothing about general conditions to compare the crash conditions to. As mentioned earlier, to solve this issue I wrote a python script (with the help of ChatGPT) to access a weather API and create my own data set of random weather points throughout Colorado. This also highlights the benefit of choosing Colorado for this project, as bounding the inputs for Colorado using latitude and longitude coordinates is straightforward since Colorado is a rectangle. The new data set will be treated as 'synthetic data,' containing information about situations where crashes theoretically did not occur. While this does not take into account how often there are crash scenarios, the data points will be able to provide an understanding of what normal weather points are which can then be compared to the weather points from the positive crash scenarios. Before running the prediction models using the weather data, I needed to do one more level of cleaning on the crash data set by removing any row with NA values. This left both sets of weather points with around 60,000 rows. My initial comparison consisted of creating several boxplots to compare the weather data from the two data sets.



The plots show me that the general range of the data points is similar, which shows me that despite the sources being different, the range of the data is still comparable. It is also important to note the range from the wind speed and precipitation data and how many outliers there are.

To test the exact relationship I ran a logistic regression model and a decision tree. The results from the two models were drastically different. Where the logistics regression was giving accuracy scores of under 60%, the decision tree was consistently yielding accuracies of over 95%. Unfortunately, the high accuracy of the decision tree combined with the low level of accuracy from the regression, leads me to believe that the logistic regression model is invalid and that the relationships are non-linear. Even though both models can handle binary outputs, the regression model relies on linear relationships, whereas the decision tree is not bound by linearity. Additionally, given the nature of the data, where weather conditions can theoretically be identical for positive and negative crash scenarios, I believe that the results from the decision trees are simply too good. I believe that the model has found a pattern to recognize the different sources of weather points rather than predicting whether or not there was a crash. To test this in the future I would like to try and use another source of synthetic weather data or find the source of the weather data from the car crash data set so the sources are identical.

Logistic Regression Results:

```
Accuracy = 0.5805070517125588
Intercept = [-1.53555869]
Predictors = ['Wind_Chill(F)', 'Humidity(%)', 'Wind_Speed(mph)', 'Precipitation(in)']
Beta Values = [[ 0.01463135  0.0127606   0.04496469 -5.76469353]]
Odds increase per unit change = [[1.01473892 1.01284236 1.04599092 0.00313636]]
[[11815 10998]
 [ 8990 15845]]
```

Decision Tree Results (each tree is a different min_sample_leafs):

```
Tree1: 0.9991745019028431, 0.9622229684351914
Tree2: 0.9592008059100067, 0.9562625923438549
Tree3: 0.9554930602193866, 0.9546675621222297
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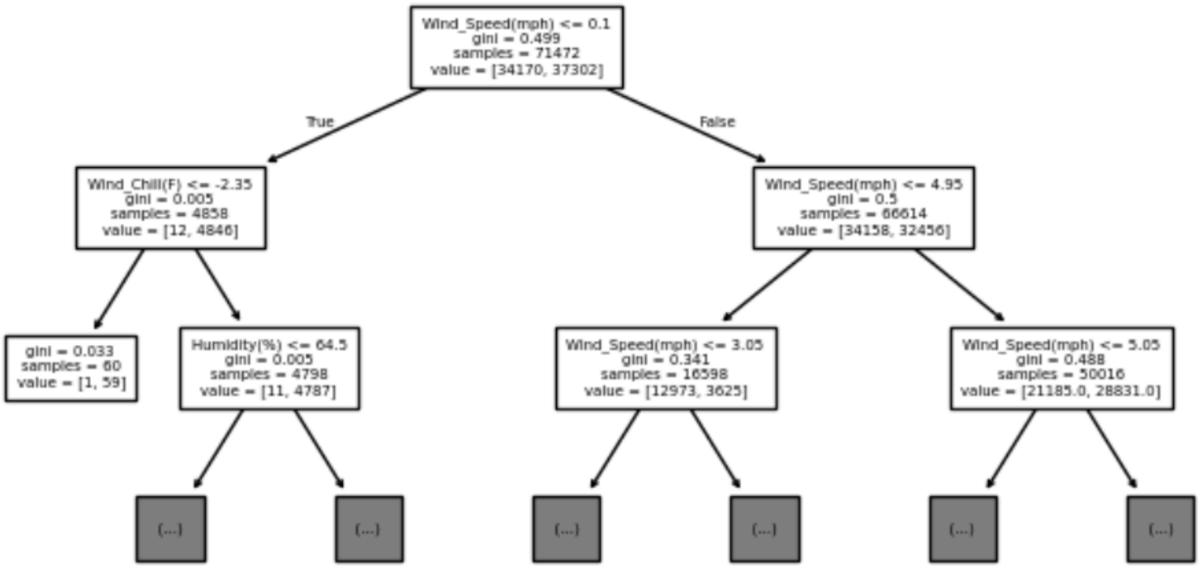

Limitations and Further Questions

The biggest limitation that pertains to all the investigations performed is that despite a correlation appearing this does not necessarily imply causation. While the way I was able to isolate the variables in the first investigation seems pretty convincing as to demonstrating causation, the correlation that I discovered in the second investigation still requires more work to determine causation. The correlation found in the second investigation could also be a result of specific city planning guides present in Colorado. Further analysis regarding schools and crashes in other states could help answer this question. The final investigation, weather conditions, has the largest limitations, as the difference in the weather conditions can possibly be a result of the data being gathered from different sources. This could be verified by putting the palace and time coordinates of the crashes into the weather API, but this is beyond the scope of this project. Additionally this study relies on the random weather points from across Colorado, not specifically where there are roads and when cars are driving. Information about how many drivers are on the road at any given time would allow me to then check the crash rate given certain weather conditions rather than crash count relative to the synthetic data. This would also allow me to say more definitively how much more dangerous certain situations are rather than just that certain situations are 'hotspots' for crashes.

General Conclusion

The overarching conclusion is that while outside factors may have an effect on how likely a crash is to occur, the best way to avoid crashes is to avoid driving when and where other people are. If possible, hitting the road before rush hour or the holiday rush can possibly help avoid crashes. Additionally, the investigation regarding proximity to schools has the potential to reveal a flaw in city planning. Depending on the results of a future study regarding crashes and schools in other states, it is possible that the correlation is not present. If this is true, it would need to be explored further as to why there is a correlation in Colorado and new plans should be designed to separate schools and locations with high amounts of crashes.